How Experience Shapes Causal World Models

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When can descriptions substitute for experience in learning to reason about a world, and what experience is needed when they cannot? I propose to compare neural systems learning from textual descriptions, observations, and action outcomes in controlled environments. The central objective is to explain which learning conditions produce representations and transition computations that support new goals and changed environmental rules. By connecting internal failures to targeted additions of evidence or supervision, the project would investigate how to allocate training experience more efficiently and build world models that support reliable planning beyond familiar trajectories.

Motivation and research gap

A model may accurately predict familiar episodes yet fail when an action has an unfamiliar consequence. Understanding that failure requires separating what the model knows about the current state from how it computes possible futures. Language can convey abstract rules economically, while action outcomes can distinguish hypotheses that ordinary observations leave unresolved. Neither source guarantees that the learner will acquire or use the relevant structure.

Vafa et al. show that generative models can pass common diagnostics while lacking coherent world models [1]. Dreamer demonstrates the utility of learned dynamics for behavior learning [2], and causal world-model interfaces already support language-based planning [3]. Recent work combines visual simulation with abstract reasoning [4]; an intervention audit further shows that accurate reward prediction need not imply faithful imagined action effects [5]. Representation-alignment results for JEPAs concern conditional representational correspondence, not equivalence of action-conditioned reasoning [6]. I would investigate how the source and organization of training evidence produce these distinctions, and whether diagnosing them predicts which additional experience repairs transfer.

Research questions

Under matched access to environmental information, how do learning from descriptions and learning predictive dynamics differ in the world variables they retain and the computations they use? Which additional evidence or supervision makes those computations reliable under new goals or changed mechanisms? A world model here means a learned state representation together with a computation of action-dependent outcomes, whether implicit in a language model or explicit in a latent predictor. Reliability concerns fidelity to the environment, not agreement between models.

Proposed approach

Build paired learning environments. I would begin with a small object-based world containing locations, switches, and conditional action effects, with known causal dependencies. The same episodes would be rendered as faithful text and structured observations. These routes would contain the same observed variables and action outcomes; text would not be equated with passive experience. Starting with structured observations would separate reasoning from perception. A subsequent visual rendering would test whether the account survives perceptual demands.

Separate information from learning design. The initial comparison would cross text versus object-state inputs with next-observation prediction versus action-conditioned latent prediction, using comparable model capacity and a shared transition backbone where possible. Both objectives would receive action information. Full language-model and recurrent world-model systems would provide a later test of external validity, with architecture and pretraining differences explicitly acknowledged. Separate conditions would add abstract rule descriptions or actively selected interventions. These supply different evidence, so their benefit would be measured against episode descriptions conveying the same causal facts, rather than attributed to modality alone.

Test transfer that requires understanding possibilities. Evaluation would cover unseen goals, unfamiliar combinations of known mechanisms, and local rule changes. After a rule change, all models would receive informative observations or descriptions before being asked to adapt. Some environments would deliberately admit multiple causal explanations under the initial observational data. The reference solver would identify when predictions are underdetermined; uncertainty would be appropriate until a diagnostic intervention or rule statement resolves the ambiguity. This respects the identifiability issues central to causal representation learning [7]. I would measure multi-step prediction, planning success, and adaptation per additional informative example.

Locate the computational source of failure. At matched states, probes would test whether action-relevant variables are available internally. Interventions using states from minimally different episodes would then test whether those variables influence predicted consequences and subsequent choices appropriately. For example, a model might encode that a switch is disabled but still predict that activating it opens a door. Comparing state access, transition fidelity, and goal-conditioned choice would distinguish lost information from an incorrect transition or planning computation. Unrelated-variable controls and held-out intervention combinations would challenge the account. Models would be compared by their causal responses in the common environment; similar internal geometry would not be required.

Test which experience repairs the failure. From the same training checkpoint, I would compare a targeted intervention episode, its faithful textual description, an explicit rule explanation, and random or difficulty-matched extra experience. The diagnosis would predict which missing distinction each addition should establish and when its effect should transfer to new goals. Equal-evidence comparisons would assess learning efficiency; separate data-budget curves would quantify the value of access to richer evidence. Where a planner is modular, a shared planner and an oracle-dynamics control would help isolate world-model error. Success requires improved internal use of the relevant variable and better transfer, not merely better reconstruction.

Feasibility and expected contribution

The first three months would establish paired data, reference solvers, and small-model baselines. Months four to eight would test one recurring transfer failure and its internal explanation. Months nine to twelve would evaluate targeted experience and generalization to a second set of mechanisms. The initial project requires neither a large video foundation model nor a robot; those settings would follow only after a controlled result.

The contribution would be an explanation of when descriptions and predictive experience provide substitutable or complementary routes to useful world knowledge. It would connect a model's computational failure to a concrete decision about what to teach it next, with potential benefits for data-efficient planning and adaptation. Longer-term work could compare human learning from instructions and interaction on the same tasks, then test specific hypotheses about rule updating or simulation against neural data with appropriate collaborators. Similar behavioral success would not, by itself, establish a shared mechanism.

Selected references

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